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UTM
UNIVERSITI TEKNOLOGI MALAYSIA

FACULTY OF COMPUTING
UTM Johor Bahru

COURSE CODE : SECD2613
COURSE NAME : SYSTEM ANALYSIS AND DESIGN
YEAR / PROGRAM : 1 / 2 (SECB/SECJ/SECR/SECV)
TIME : 07 FEBRUARY 2023
DATE :
PLACE :

INSTRUCTIONS TO CANDIDATES:

1. Fill in your particulars in the question booklet and answering booklet:

Name	
Passport No.	
Matric No.	
Year / Program	
Lecturer	

2. Read instructions given for each question carefully.
3. Answer **ALL** questions.
4. This paper consists of three sections.

Section A: **10 MULTIPLE CHOICE** questions (10 Marks).

Section B: **4 STRUCTURED** questions (60 Marks).

Section C: **1 CASE STUDY** question (30 Marks).

TOTAL MARKS is **100 Marks**

THIS PAPER CONSISTS OF 13 (THIRTEEN) PRINTED PAGES INCLUDING THIS PAGE.

SECTION A : MULTIPLE CHOICE QUESTIONS
10 MARKS (EACH QUESTION 1 MARK)

INSTRUCTION : ANSWER ALL QUESTIONS.

Please write your answer (A/B/C/D) in the answer sheets.

Question 1

Which of the statements **BEST** describe a logical data flow diagram?

- A. Logical data flow diagram includes types of programs, such as online or batch.
- B. Logical data flow diagram is a model of how the system will be implemented. *physical*
- C. Logical data flow diagram does not include any primitive processes.
- ☒ D. Logical data flow diagram is a model of how the business operates.

Question 2

X is use when not every condition is relevant to every action.

Based on the above statement, what is X?

- A. Structured English
- B. Decision Table
- C. Structured Tree
- ☒ D. Decision Tree

Question 3

Which of the following is the purpose of a (control couple) in a structure chart?

- ☒ A. Pass parameters from a subordinate module to the control module.
- B. Pass parameters from the control modules to a subordinate module.
- C. Pass data from a subordinate module to the control module.
- D. Pass data from the control module to a subordinate module.

Question 4

Which of the following is **NOT** a form of electronic output?

- A. Fax
- B. Email
- ☒ C. Report
- D. Bulletin Board Message

Question 5

Which of the following is a **FALSE** statement regarding the purpose of system implementation?

- A. To convert final physical system specifications into working and reliable software.
- B. To document work that has been done.
- C. To provide help and support for current and future users.
- ☒ D. To modify a system to correct faults, to improve performance, or to adapt the system to a changed environment or changed requirements

Question 6

Which of the following is **FALSE** about the (user interface design?)

- A. Availability of operating systems will be supported.
- ☒ B. Minimize the usability and interaction between devices and users.
- C. Types of devices will be used.
- D. The system is designed to use a browser.

Question 7

Which of the following is **FALSE** about the (decision table?)

- A. Effectively avoids impossible situations
- B. Effectively avoids redundancies
- ☒ C. Effectively avoids miscommunication to end users
- D. Effectively avoids contradictions

Question 8

Consider the transaction below:

Transaction 1: add new staff

Transaction 2: change a venue of an event

Transaction 3: change a customer phone number

Transaction 4: show a list of events in 2020

Transaction 5: change a staff bank account number

Which of the following CRUD matrix will represent the above transaction?

A.

Activity	CUSTOMER	EVENT	STAFF
Transaction 1			C
Transaction 2		U	
Transaction 3	C		
Transaction 4		R	
Transaction 5			D

B.

Activity	CUSTOMER	EVENT	STAFF
Transaction 1			C
Transaction 2		C	
Transaction 3	C		
Transaction 4			R
Transaction 5			D

C.

Activity	CUSTOMER	EVENT	STAFF
Transaction 1			C
Transaction 2		U	
Transaction 3	U		
Transaction 4		R	
Transaction 5			U

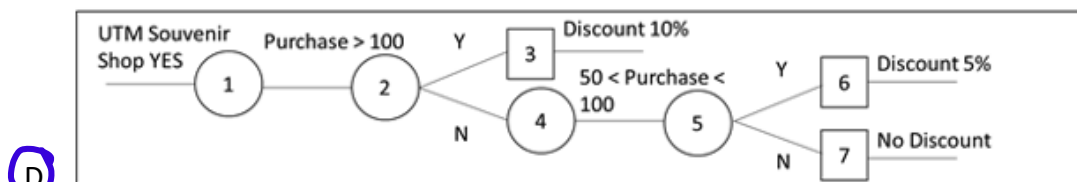
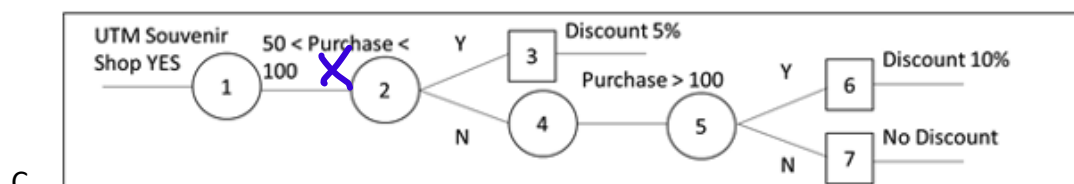
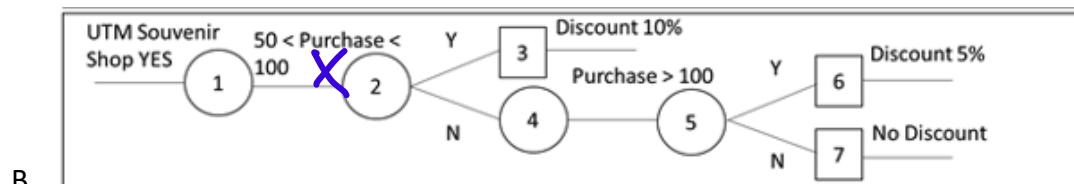
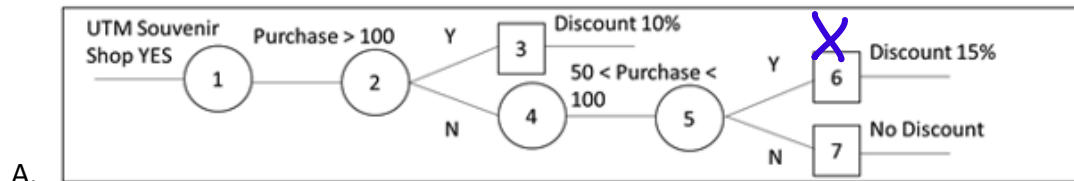
D.

Activity	CUSTOMER	EVENT	STAFF
Transaction 1			C
Transaction 2		C	
Transaction 3	U		
Transaction 4		R	R
Transaction 5			D

Question 9

UTM Souvenir Shop plan to conduct Year End Sale, where if the amount of the purchase is (greater than RM100) the customer will received 10% discount, if the amount of the purchase is (between RM50 and RM100,) the customer will be received 5% discount and if the amount is less than RM 50, the customer will not receive any discount.

Which one of the decision trees below is TRUE to represent the above scenario?



Question 10

XYZ University decided to change the registration information system and to replace it with a new system. The new registration system was developed internally by the information technology staff. Now, the information technology staff want to install the new registration system. The manager of the registration department requires that the installation involves comparing the results of the new system with the old one.

Based on the above scenario, which of the following installation strategies is the **MOST** appropriate?

- A. Direct Installation
- ☒ B. Parallel Installation
- C. Single-location Installation
- D. Phased Installation

SECTION B : STRUCTURED QUESTIONS

60 MARKS

INSTRUCTION : ANSWER ALL QUESTIONS.

Please write your answer in the answer sheets.

QUESTION 1 (15 Marks)

Figure 1 shows Logical DFD Level 1 for one of the processes in Cinematics Ticketing System.

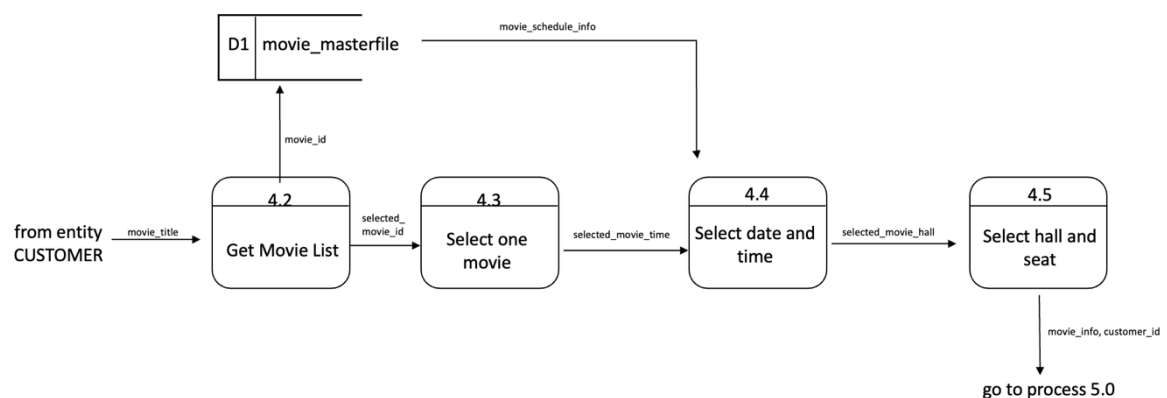


Figure 1: Logical DFD Level 1 (Process 4.0 CHOOSE MOVIE)

A detailed flow for *Process 4.0 Choose movie* is explained as below:

The process starts when the customer goes to the movie section on the website. The list is retrieved from the movie_masterfile and a complete list was shown to the customer. Based on the list, the customer can choose one movie that they are interested in. The selected movie will get the schedule data from a temporary schedule file and the customer is allowed to choose the date and time that they preferred. Then the hall that has the movie show that was selected will be displayed so the customer can select the seating that they interested. The hall and seating view will also display the ticket price. Note that different seating would have different price, and this is kept in another temporary file for the ticket price. After the seat was chosen, the total price is displayed, and the customer will be taken to the payment page (Process 5.0). This is the end for Process 4.0.

- a) Identify the main processes from the description that can assist you to convert Figure 1 to a PHYSICAL DFD.

(Hint: It can be based on Figure 1 and other process you can identified from the description)

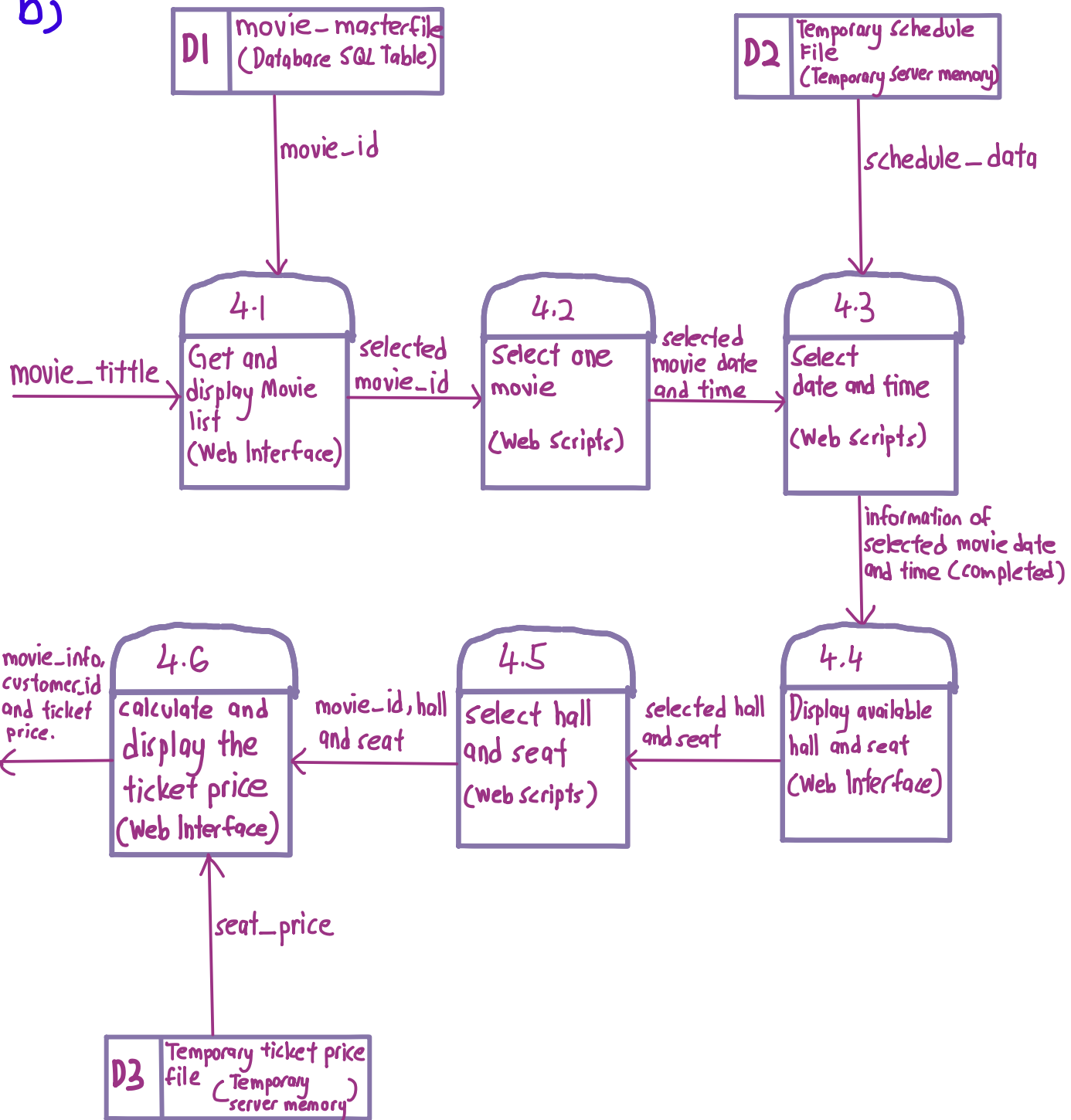
[6 Marks]

- b) Draw a PHYSICAL DFD Level 1 for *Process 4.0 Choose movie* based on your answer in (a).

[9 Marks]

- 9) - Get and display movie list
- select one movie
 - select date and time
 - display available hall and seat
 - select hall and seat
 - calculate and display the total price

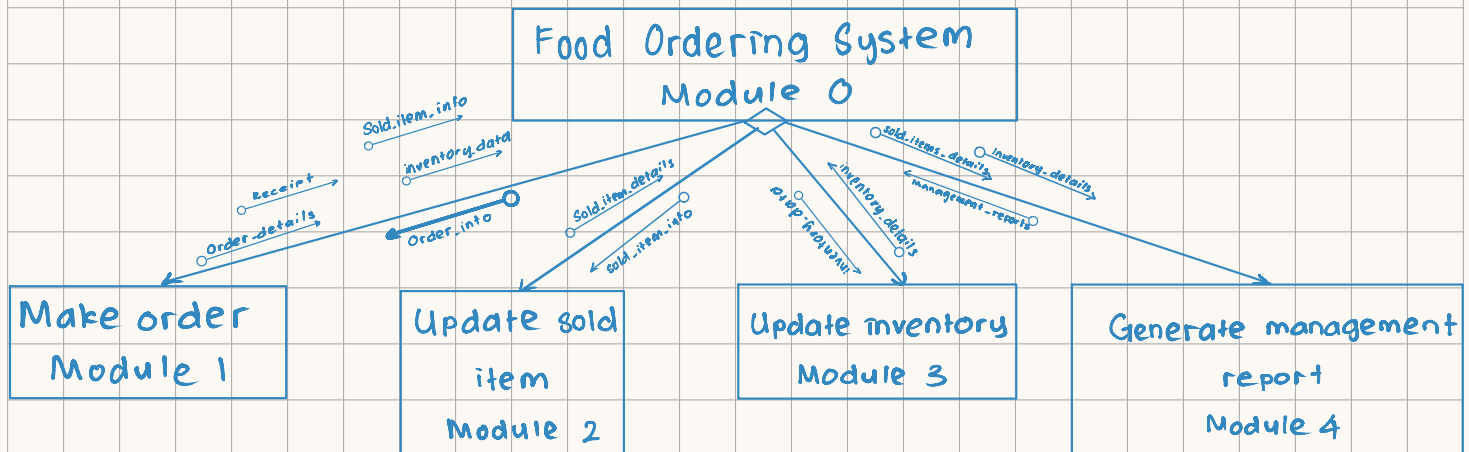
b)



Question 2

a. Structure chart is a top-down representation of business function and processes that show the program modules and the relationship among them.

b.



c.

Module 2 has a high degree of collision and is loosely coupled.

Because it perform a single function and not affecting the other modules

d. STAR1 Manage Order Process

Repeat

Call Collect Order

Call Verify Order

Call Make Payment

END Repeat

Call Generate Order List

END STAR1

d) Write the Structured English for the structure chart in Figure 3.

(4 Marks)

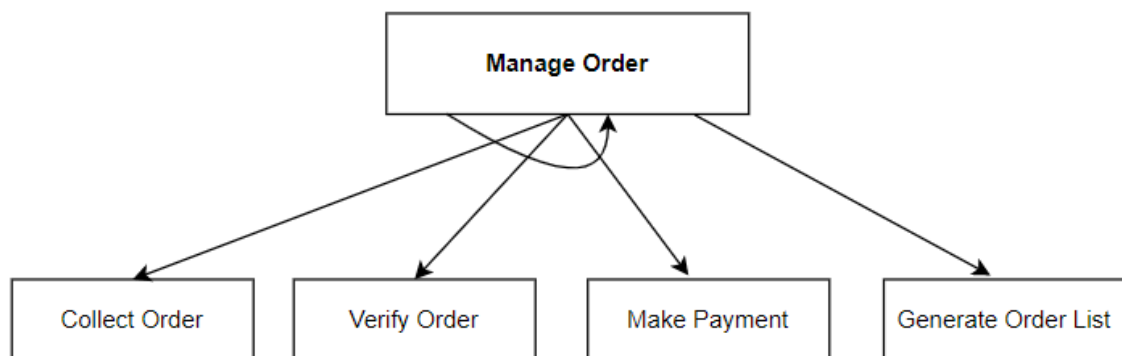


Figure 3: Structured chart for Manage Order Process

QUESTION 3 (15 Marks)

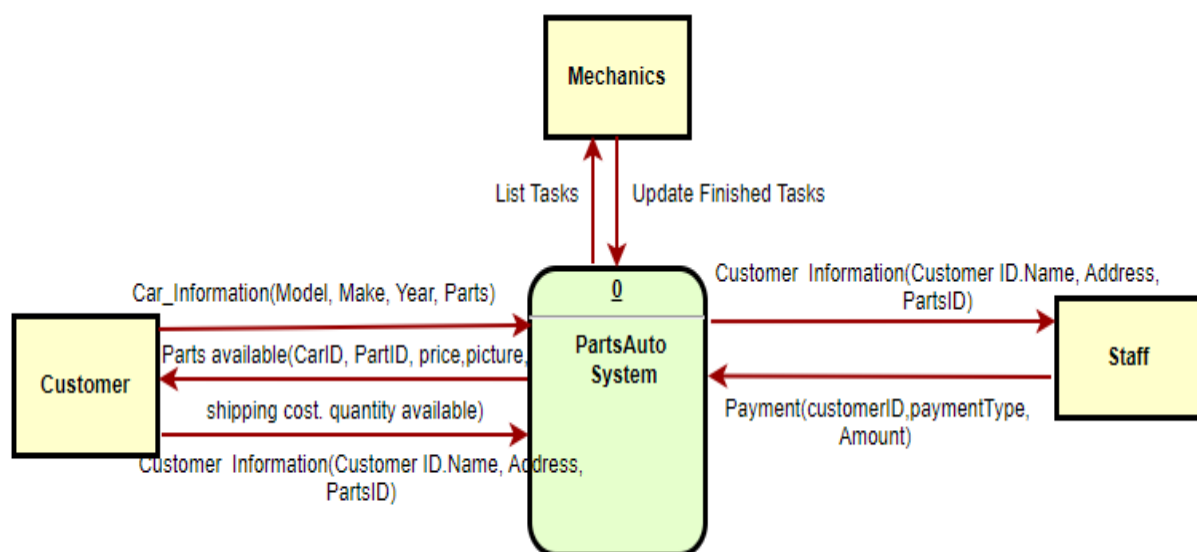


Figure 4: Context Diagram

PartsAuto is a company selling recycling parts for classic and antique cars. Customer can search for information related to their car and find available parts for sale. Mechanics can be assigned task for installing the parts in the customer's car and update their finished task while staff will manage the customers history of purchased parts and their payments.

- a) Based on the Context Diagram shown in Figure 4 identify **THREE** input and **THREE** output screens to be designed. **(6 Marks)**
- b) You have been given the task to design a Web-based display for Parts Auto Company where:
- Customer can search for make, model, and year. When the data of the car is found, it will show all the available parts of the car.
 - Information for the car parts are: the stock is in or not; the part description; condition of the part; price, and shipping cost are displayed; how many quantity available for each part, along with a picture of the part, button for each part that may be clicked to purchase the part.

Figure 5 shows the initial design which really needs improvements. Improve the GUI design by drawing the GUI with suitable text boxes, varieties of GUI such as drop-down menus, check boxes, and radio buttons which are suitable to create interest in the form. **(9 Marks)**

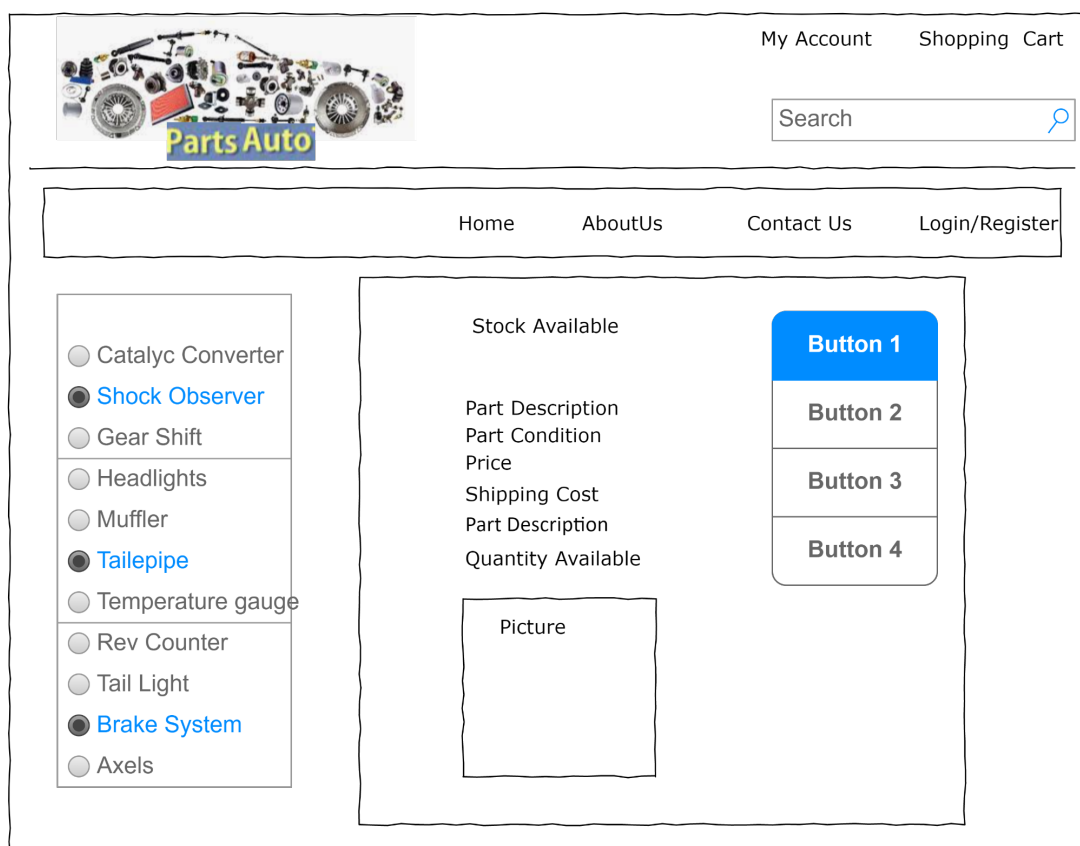


Figure 5: Initial GUI Design

SEARCH FOR PARTS

Make: [▼ Select Make] Model: [▼ Select Model] Year: [▼ Select Year]

[🔍 Search Parts]

SEARCH RESULTS(For: 1967 Ford Mustang)

Image	Part Details
[Picture 1]	**Part #:** ENG-7845 **Description:** 289 V8 Engine Block (Rebuilt) **Condition:** ☒ New ☐ Refurbished ☐ Used **Stock:** ☒ In Stock ☐ Out of Stock **Price:** \$1,200.00 **Shipping:** \$75.00 **Qty Available:** 3 [Add to Cart] [Add to Wishlist]
[Picture 2]	**Part #:** BRK-3210 **Description:** Disc Brake Caliper Set (Front) **Condition:** ☒ New ☐ Refurbished ☐ Used **Stock:** ☐ In Stock ☒ Out of Stock **Price:** \$350.00 **Shipping:** \$25.00 **Qty Available:** 0 [Notify Me When Available]
[Picture 3]	**Part #:** SUS-9981 **Description:** Front Coil Spring Set (Standard) **Condition:** ☐ New ☒ Refurbished ☐ Used **Stock:** ☒ In Stock ☐ Out of Stock **Price:** \$85.00 **Shipping:** \$15.00 **Qty Available:** 5 [Add to Cart] [Add to Wishlist]
[< Previous Page] [1] [2] [3] [Next Page >]	
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Input

Customer Search Screen: To input criteria (e.g., make, model, year) for searching car parts.

Mechanic Task Update Screen: For mechanics to log in and update the status of their assigned installation tasks (e.g., "in progress," "completed").

Payment Processing Screen: For staff to record and process customer payments for parts and services.

Output

Available Parts Display Screen: To show the results of a customer's search, displaying part details, stock status, prices, and pictures.

Customer Purchase History Screen: For staff to view a customer's complete history of purchased parts and payments.

Mechanic Task Assignment Screen: For staff to view a list of tasks and assign them to specific mechanics.

Question 4

- a) 1. Testing
2. Installation
3. Training
- b) 1. User Acceptance Testing, testing is based on defined user requirements.
2. Alpha testing, tests performed at developer's site by computer
3. Beta testing, ~~and~~ conducted at the customer site end user or computer
- c) i. Parallel Installation, low risk because the old system remains available if the new system fail.
ii. Phased Installation, easier to manage since the new system is introduced gradually, allowing problems to be fixed step by step
iii. Direct installation, employees only use the new system, avoiding the need to enter the same data into both old and new systems.

SECTION C : CASE STUDY

30 MARKS

INSTRUCTION : ANSWER ALL QUESTIONS.

Please write your answer in the answer sheets.

In early 2019, GigaBites - a computer supplies company has set up number of new branches for their business operation in Johor. GigaBites sends out invoices monthly and will offers discounts if the clients made payments within 18 days after purchases. Due to COVID-19 pandemic that disrupts the economy globally, GigaBites decides to revise the discounting policy for the clients to sustain their business environment.

The revised discounting policy is as follows: If the amount of the order for computer supplies is greater than \$1,000, subtract 4 percent for the order; If the amount is between \$500 and \$1,000, subtract a 2 percent discount; If the amount is less than \$500, do not apply any discount; All orders made via the Web automatically receive an extra 5 percent discount; Any special goods order (computer tables, for example) is exempt from all discounting.

- a) Propose the structured English to solve the GigaBites company situation for the revised discounting policy. **(6 Marks)**
- b) Develop a simplified decision table for GigaBites discounting decisions, for which the condition alternatives are limited to Y and N. **(8 Marks)**
- c) Draw a decision tree to illustrate the possible outcomes for the revised discounting decisions. **(6 Marks)**
- d) Sketch the graphical user interface (GUI) design for the GigaBites company manager to overview the sales summary report using their inventory management system. **(8 Marks)**

The GUI design should fulfill the following needs highlighted by the manager during interview:

"This report should give me an overview of GigaBites sales for a given time period. The generated sales report should also show the cost of goods sold, gross profit, margins, and

tax, so I can have a better idea of how much money is actually going back to GigaBites business.

Also, the sales summary report provides a more macro view of GigaBites retail sales: How much did company make last month or the previous month? What about company year-to-date sales compared to last year? These numbers should provide some general insights around the health of GigaBites business, and it can inform GigaBites medium to long-term decisions."

- e) Justify the use of appropriate output elements that are included in the GUI design in Question d. (2 Marks)

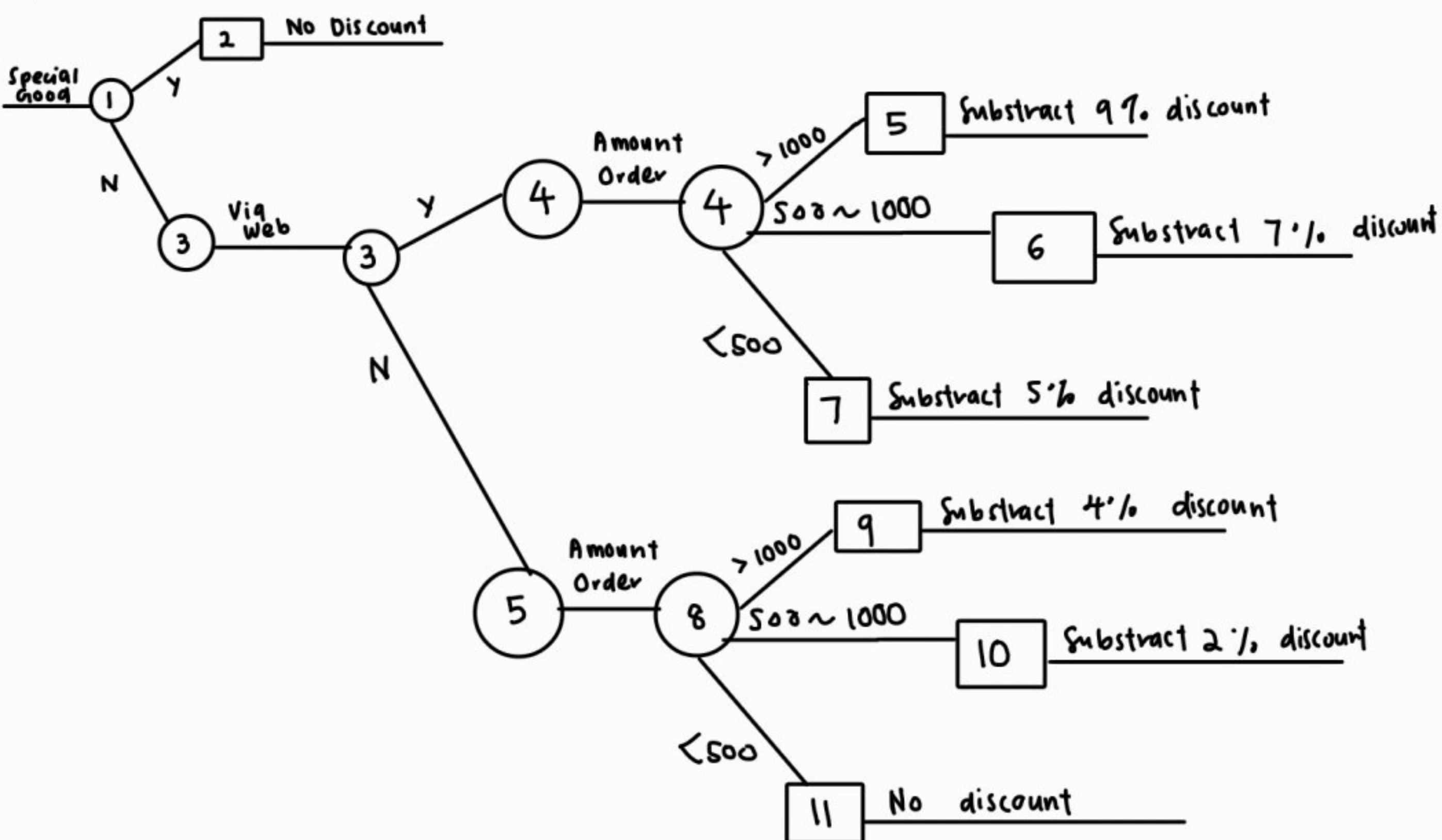
Answer :

```
a) START
  IF Order Item is Special Goods
    Apply no discount
  ELSE
    IF Order Amount > 1000
      Discount 1 = 4%
    ELSE IF Order Amount ≥ 500 && ≤ 1000
      Discount 1 = 2%
    ELSE
      Discount 1 = 0%
    END IF
  END IF
  IF Ordering Channel is Web
    Discount 2 = 5%
  ELSE
    Discount 2 = 0%
  END IF
  Calculate Price = Order Amount * (100% - Discount 1 + Discount 2)
STOP
```

b)

Condition	Rules						
	1	2	3	4	5	6	7
Special Goods Order	Y	N	N	N	N	N	N
Amount Order > 1000	-	Y	Y	N	N	N	N
Amount Order ≥ 500 & ≤ 1000	-	-	-	Y	Y	N	N
Ordering channel = Web	-	Y	N	Y	N	Y	N
Actions							
Subtract 9% discount		X					
Subtract 7% discount				X			
Subtract 5% discount						X	
Subtract 4% discount			X				
Subtract 2% discount					X		
No Discount	X						X

c)



d)

×

SALES SUMMARY REPORT

From : To :

Financial Overview Period Metrics

Gross Sales Revenue:

Gross Profit Margins:

Net Profit Achieved:

Cost of Good Sold (COGS):

Total Tax Obligations:

HISTORICAL COMPARATIVE DATA ANALYSIS

2,XXX

2,XXX

2,XXX

EXPORT CSV

EXPORT PDF

• This month Vs Last Month : - 8.9 %

• Year to Date Revenue : XX k Last Revenue : XX k

Status :

Not Healthy

Healthy

e)

- 1- Visual Bar Chart Component which provide instant view and idea on month to month and year to year trend.
- 2- Export Icon so user can directly download all the report that has been generated.